

Practice Question Set

1. What is a constant?
2. Differentiate between a variable and a constant.
3. What are the two broad classifications of constants in C?
4. What are primary constants?
5. What are integer constants? Give examples.
6. What are real (floating-point) constants? Give examples.
7. What are character constants? Give examples.
8. What is a string constant?
9. Why is the null character ('\0') added to strings?
10. What is a pointer?
11. What is a structure?
12. What is a union?
13. Differentiate between structure and union.
14. Why are unions memory efficient?
15. Differentiate between integer, real, and character constants.

Important Interview/Viva Questions

16. Why is a constant called immutable?

17. Can the value of a constant be modified during program execution?
18. Difference between variables and constants.
19. Difference between string constants and character constants.
20. Difference between structure and union.
21. Why is '\0' important in strings?
22. Which data structure uses shared memory?
23. Which data structure allocates separate memory for each member?
24. Why are unions more memory efficient than structures?
25. Give real-life examples of integer, floating-point, and character constants.

Answers

1. What is a constant?

Answer:

A constant is a fixed value that cannot be changed during program execution.

Example:

```
const float PI = 3.14;
```

2. Differentiate between a variable and a constant.

Variable	Constant
Value can change	Value cannot change

Variable	Constant
Mutable	Immutable
Reassignment allowed	Reassignment not allowed
Example: age=20; age=21; Example: const float PI=3.14;	

3. What are the two broad classifications of constants?

Answer:

1. Primary Constants
 2. Secondary Constants
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4. What are primary constants?

Answer:

Primary constants represent basic and individual data values.

Types:

- Integer constants
 - Real constants
 - Character constants
-

5. What are integer constants?

Answer:

Integer constants are whole numbers without decimal points.

Examples:

10
-15
0
500

6. What are real constants?

Answer:

Real constants are numbers containing decimal points.

Examples:

3.14
-12.35
0.75

7. What are character constants?

Answer:

Character constants are single characters enclosed within single quotes.

Examples:

'A'
'7'
'@'
'\$'

8. What is a string constant?

Answer:

A string constant is a sequence of characters enclosed within double quotes.

Example:

"Hello"
"Gaurav"

C automatically adds a null character ('\0') at the end of every string.

9. Why is the null character ('\0') added to strings?

Answer:

The null character indicates the end of the string and helps the compiler identify where the string terminates.

10. What is a pointer?

Answer:

A pointer is a variable that stores the memory address of another variable.

Example:

```
int x = 10;  
int *ptr = &x;
```

11. What is a structure?

Answer:

A structure is a user-defined data type that groups different types of data under a single name.

Example:

```
struct Student  
{  
    int age;  
    float marks;  
    char grade;  
};
```

12. What is a union?

Answer:

A union is similar to a structure, but all members share the same memory location.

Example:

```
union Data  
{  
    int i;  
    float f;  
    char c;  
};
```

13. Differentiate between Structure and Union.

Structure	Union
Separate memory for each member	Shared memory for all members
All members can store values simultaneously	Only one member stores value at a time
Uses more memory	Uses less memory

14. Why are unions memory efficient?

Answer:

Because all members share the same memory location, resulting in less memory usage.

15. Differentiate between Integer, Real, and Character Constants.

Integer Constant	Real Constant	Character Constant
No decimal point	Contains decimal point	Single character
Example: 100	Example: 3.14	Example: 'A'
Whole numbers	Fractional numbers	Enclosed in single quotes

Important Interview Answers

16. Why is a constant called immutable?

Answer:

Because its value remains fixed throughout program execution and cannot be modified.

17. Can the value of a constant be modified?

Answer:

No.

Example:

```
const float PI = 3.14;
```

```
PI = 5.0; // Error
```

18. Difference between Variables and Constants.

Variable	Constant
Value changes	Value remains fixed
Mutable	Immutable
Reassignment allowed	Reassignment prohibited

19. Difference between String Constants and Character Constants.

String Constant	Character Constant
Enclosed in double quotes	Enclosed in single quotes
Sequence of characters	Single character
Example: "Hello"	Example: 'H'

20. Difference between Structure and Union.

Answer:

- Structures allocate separate memory for each member.
 - Unions share the same memory among all members.
 - Structures allow all members to contain values simultaneously.
 - Unions allow only one member to contain a value at a time.
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21. Why is '\0' important in strings?

Answer:

It marks the end of the string and helps string functions determine where the string ends.

22. Which data structure uses shared memory?

Answer:

Union uses shared memory.

23. Which data structure allocates separate memory for each member?

Answer:

Structure allocates separate memory for each member.

24. Why are unions more memory efficient than structures?

Answer:

Because all members occupy the same memory location, reducing memory consumption.

25. Give examples of integer, real, and character constants.

Answer:

Integer Constants

10
-20
100

Real Constants

3.14
0.75
-5.67

Character Constants

'A'
'5'
'@'